

**Multi Light Band Analysis of
Concentration, Asymmetry, and Smoothness (CAS) of E+A Galaxies
with Data from the Vera C. Rubin Observatory's
Legacy Survey of Space and Time (LSST)**

Isabella Troy Brazoban, Lorik Fazliu, and Ali Philip

Advised by Dr. Charles Liu

The City University of New York

College of Staten Island

Department of Physics and Astronomy

2800 Victory Blvd, Staten Island, NY, 10314

Abstract

The study of galaxy morphology has historically been constrained by observational limits. The Vera C. Rubin Observatory's Legacy Survey of Space and Time (LSST), generating approximately 20 terabytes of raw astronomical data per night, presents an unprecedented opportunity to resolve this data bottleneck. This ongoing preliminary research develops an automated morphological classification pipeline within the Rubin Science Platform to efficiently identify post-starburst (E+A) merger remnants. Utilizing Jupyter notebooks, we processed photometric data across six optical and near-infrared light bands (u, g, r, i, z, y). To manage this massive data volume, the methodology prioritizes scalable software architecture. SQL queries isolated the target galaxies to a localized database, enabling time-efficient, iterative querying without requiring repeated raw data pulls. The initial query included 100,000 objects which were filtered using specific magnitudes. Filters included the u-band (u) where $(15.0 \leq u \leq 21.0)$, the differences of the u and g-band (g) where $(1.0 \leq u - g \leq 1.8)$, and the g and r-band (r) where $(0.3 \leq g - r \leq 0.85)$. The isolation of this transition zone (known as the "Green Valley") yielded a refined sample of 103 target galaxies (out of 100,000). While the Conselice (2003) Concentration, Asymmetry, and Smoothness (CAS) framework is well-established, applying it at the LSST scale requires novel computational infrastructure. To automate the Conselice framework, the CAS parameters were manually engineered using composite model fluxes and second-order shape moments. Concentration was derived from flux-weighted radii, asymmetry was computed through 180-degree flux array mirroring, and smoothness applied a scipy uniform filter, dynamically scaled to 0.3 of the approximated Petrosian radius to isolate structural clumps while preventing null processing errors. The establishment of this automated foundation is a step toward conducting large-scale analyses of cosmic merger rates and understanding how mass, age, and underlying structure dictate galactic evolution.